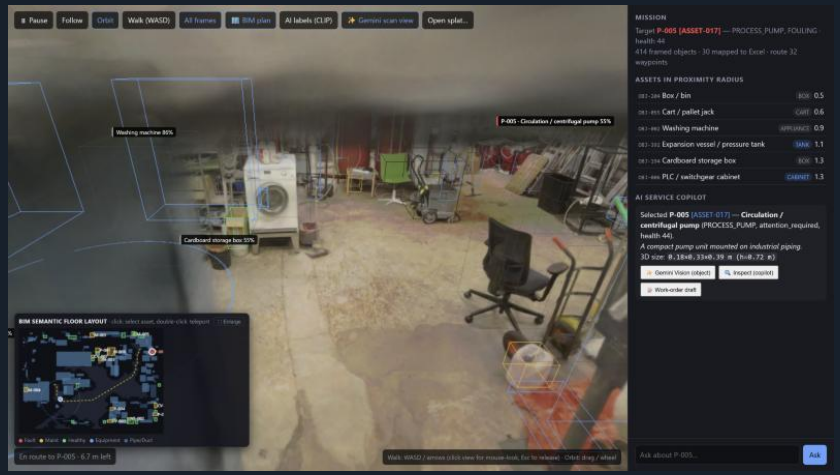
Semantic digital twin and field-service copilot

theMOLFARS · Valmet × Since AI 2026 · Industrial Metaverse challenge



414

3D-framed objects, no overlaps

30 / 30

Excel assets mapped

8.9 m

collision-free A* route

49 / 50

vision benchmark crops

From blind scan to navigable twin

1

Sanitize

1.68 M splats read; corrupted tails cut

2

Floor + grid

Floor/ceiling peaks, 10 cm occupancy grid

3

Segment

Colour-aware voxel graph + 3D NMS

4

Recognise

CLIP + video photo match + Gemini

5

Map Excel

Fit score: class, size, vision

6

Route

A* with clearance → scene.json

pipeline/ (Python)

build_scene.py runs offline in ~5 s and writes scene.json, the clean splat and a floor plan

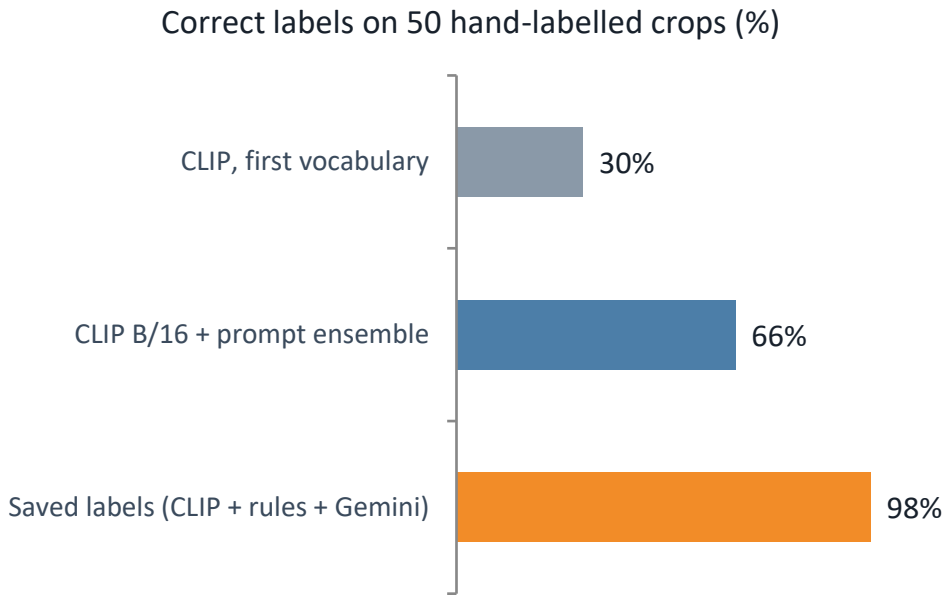
web/ (three.js + Spark)

Renders the splat, frames, avatar, proximity list and copilot chat; CLIP runs on WebGPU

server.py (FastAPI)

PostgreSQL evidence, Gemini answers, vision labels, work-order drafts

414 non-overlapping, named 3D frames



49/50 is an upper bound for this room: part of the rules were written for this scan.

Colour-aware segmentation

4 cm voxels join only if their colours match, so a red pump splits from its white pipe

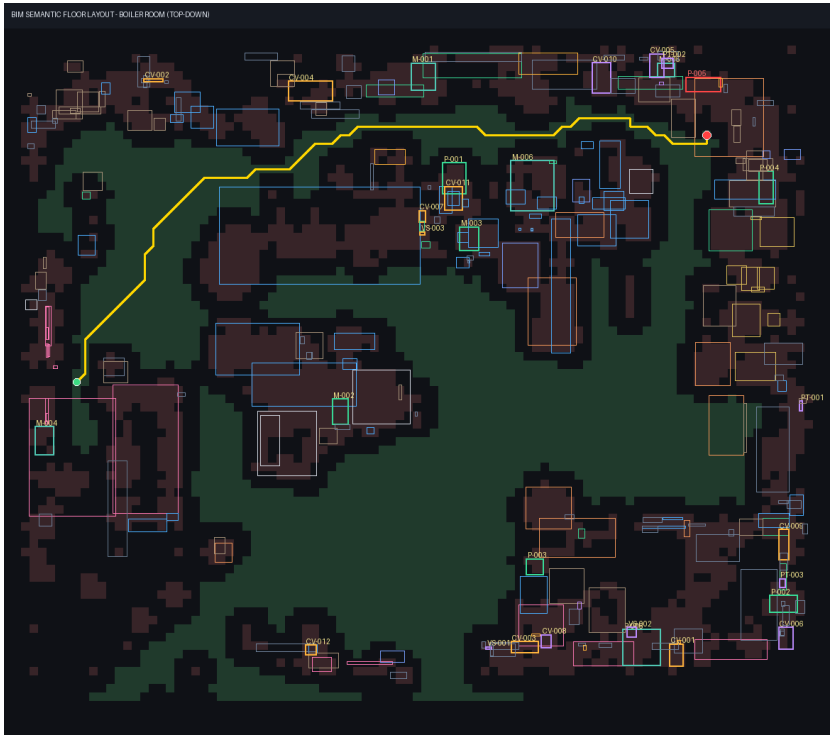
One frame per object

3D NMS, 129 touching parts grouped into 49 objects, 88 frames rotated by PCA; isolate any of them

17 classes, honest names

Pumps, valves, gauges, motors — and the washing machine, boxes and ladder

A route that never clips through equipment



Occupancy grid, 10 cm

Floor = splats within ±12 cm; obstacles 0.2–1.7 m high; every cell 0.22 m from the nearest obstacle splat

A* with clearance cost

Middle of aisles, line-of-sight smoothing; Walk mode collides with the same grid

Proximity radius 1.3 m

Frames and Excel status appear as the avatar passes: 14 assets plus the target on this route

Inspection tour (Q-04)

13 non-healthy assets, urgent first, then the shortest walk: 57 m

Excel data lands on boxes that fit

Worst health is the target

ASSET-017 / P-005, process pump, fouling, health 44, on box OBJ-115 that vision reads as a pump

Fit score per asset

Exact class +4, compatible class +1.5, seen by vision +2, size check per class

Never on walls or clutter

Boxes read as wall, pipe, box, chair or appliance are excluded; a test fails if that breaks

Some of the 14 assets the avatar passes within 1.3 m

Tag	Class	State	Health
P-005	Process pump	attention required	44
M-006	Electric motor	attention required	45
P-001	Process pump	degraded	69
M-001	Electric motor	watch	77
CV-010	Control valve	healthy	96
PT-002	Pressure transmitter	healthy	96

Statuses are the Excel values as they are: the route also passes degraded and watch-listed devices.

Cited diagnosis, honest confidence



Arrival at P-005



SQL over 8 Excel tables



Gemini or offline template



4-section cited report

- Comments on every asset the avatar passes, with evidence IDs
- Every cited ID is checked against the retrieved data (Q-03)
- Own telemetry detector: AUC 0.91 vs the pack's labels (Q-02)
- Work-order draft, never submitted (Q-05)

Confidence for P-005 (Q-06)

0.91	model confidence from the pack
0.43	after 9 open data-quality flags
0.55	3D identity of the mapped box
0.43	overall: the weakest link

Modular backend that is safe to demo

Pipeline

- build_scene.py: splat → scene.json
- extract_frames.py: video frames
- fetch_data.py: splat from Google Drive
- video_lift.py: COLMAP + CLIP-L + OWLv2 on GPU

API (pack's API_Contract)

- Every API_Contract route of the pack
- POST /api/telemetry + MQTT bridge (Q-07)
- OpenUSD export of the scene (Q-09)
- /api/priorities, /anomalies, /brief

Trust and reliability

- docker compose up runs everything
- CI: 21 unit tests on every push
- JSON-only POSTs, escaped labels
- SQLite fallback when Docker is off

Five-minute demo, and what comes next

1. Orbit overview: frames over the splat
2. Floor plan: grid, obstacles, A* route
3. Mission: frames + comments within 1.3 m
4. P-005: cited diagnosis, isolated pump
5. Inspection tour (🎬 Demo runs all of it)

Next

- Run video_lift.py on the GPU PC for multi-view labels
- Re-run Gemini labels without the old score floor
- Connect a real OPC UA / MQTT source
- Open the OpenUSD export in Omniverse

github.com/huziichuk/theMOLFARS